Deokki Hong

AI Engineer | Rebellions, Software SDK Team

Gyeonggi-do, Republic of Korea | +82-10-3323-4562
dk.hong@yonsei.ac.kr | GitHub | LinkedIn | Google Scholar

Research Interests

- **Efficient generative models:** LLM/VLM compression, quantization, and inference optimization.
- **Hardware-aware optimization:** neural architecture search and network–accelerator co-exploration.

Experience

Rebellions Dec. 2024 – Present
AI Engineer | Software SDK Team

SAPEON KOREA Dec. 2022 – Dec. 2024
Software Engineer | Platform Team

NAVER CLOVA Mar. 2022 – Aug. 2022
AI Research Intern | Image Vision

Education

Yonsei University Sep. 2020 – Aug. 2022
M.S. in Artificial Intelligence | Seoul, Republic of Korea

Yonsei University Mar. 2013 – Feb. 2020
Bachelor's Degree in Economics and Computer Science | Seoul, Republic of Korea

Projects

Lightweight One-Stage Object Detection Mar. 2022 – Aug. 2022
NAVER CLOVA Image Vision
Analyzed lightweight one-stage detectors and examined alternative architectures.

Open-Vocabulary Object Detection (OVD) Jun. 2022 – Aug. 2022
NAVER CLOVA Image Vision
Reviewed prior work and conducted research on open-vocabulary object detection.

High-Resolution Face Image Generation with Transformer-Based GANs 2021
Electronics and Telecommunications Research Institute (ETRI)
Researched fast-converging face generation with limited training data.

Fast Distributed Deep Neural Network Training 2020
Korea Institute of Industrial Technology (KITECH)
Investigated techniques for accelerating multi-GPU deep neural network training.

Awards

Silver Prize – The 28th Samsung Humantech Paper Award Feb. 2022
K. Choi, **D. Hong**, H. Y. Lee
Computer Science and Engineering

Publications

Enhancing Generalization in Data-Free Quantization via Mixup-Class Prompting 2025
*Jiwoong Park, Chaeun Lee, Yongseok Choi, Sein Park, **Deokki Hong**, and Jungwook Choi*
IEEE/CVF International Conference on Computer Vision Workshops (ICCVW)

DANCE++: Differentiable Accelerator/Network Co-Exploration with Hard Constraints and Data-Free Training for Real-World Scenarios 2025
*Kanghyun Choi, **Deokki Hong**, Hyeyoon Lee, Joonsang Yu, Noseong Park, Youngsok Kim, and Jinho Lee*
IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD)

Online Boundary-Free Continual Learning by Scheduled Data Prior 2023
*Hyunseo Koh, Minhyuk Seo, Jihwan Bang, Hwanjun Song, **Deokki Hong**, Seulki Park, Jung-Woo Ha, and Jonghyun Choi*
International Conference on Learning Representations (ICLR)

Enabling Hard Constraints in Differentiable Neural Network and Accelerator Co-Exploration 2022
***Deokki Hong**, Kanghyun Choi, Hyeyoon Lee, Joonsang Yu, Noseong Park, Youngsok Kim, and Jinho Lee*
ACM/IEEE Design Automation Conference (DAC)

It's All In the Teacher: Zero-Shot Quantization Brought Closer to the Teacher 2022
*Kanghyun Choi, Hyeyoon Lee, **Deokki Hong**, Joonsang Yu, Noseong Park, Youngsok Kim, and Jinho Lee*
IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) | Oral presentation

Qimera: Data-Free Quantization with Synthetic Boundary Supporting Samples 2021
*Kanghyun Choi, **Deokki Hong**, Noseong Park, Youngsok Kim, and Jinho Lee*
Conference on Neural Information Processing Systems (NeurIPS)

DANCE: Differentiable Accelerator/Network Co-Exploration 2021
Kanghyun Choi, **Deokki Hong***, Hojae Yoon*, Joonsang Yu, Youngsok Kim, and Jinho Lee*
ACM/IEEE Design Automation Conference (DAC)

* Equal contribution (co-first authors).

Patents & Patent Applications

Method and Device for Modularized Compression of Machine Learning Model Jan. 12, 2026
*Chaeun Lee and **Deokki Hong***
KR20260005662A | Published patent application

Method and Apparatus for Calibrating by Using Input-Output Distribution of Neighboring Layer May 28, 2025
*Jin Woo Son, Ji Eun Lim, **Deokki Hong**, and Won Jae Lee*
KR20250075395A | Published patent application

Artificial Neural Network and Computational Accelerator Structure Co-Exploration Apparatus and Method with Hard Constraints Oct. 2, 2024
*Jinho Lee, Noseong Park, Youngsok Kim, **Deokki Hong**, Kanghyun Choi, and Hyeyoon Lee*
KR20240144573A | Published patent application

Apparatus and Method for Non-Accessible Calibration of Training Data for Artificial Neural Network Quantization Oct. 2, 2024
*Jinho Lee, Kanghyun Choi, Noseong Park, Youngsok Kim, **Deokki Hong**, and Hyeyoon Lee*
KR20240144572A | Published patent application

Artificial Neural Network and Computational Accelerator Structure Co-Exploration Apparatus and Method Dec. 6, 2023
*Jinho Lee, Youngsok Kim, Kanghyun Choi, and **Deokki Hong***
KR102610429B1 | Granted patent

Academic Service

Conference reviewer

IEEE/CVF International Conference on Computer Vision (ICCV)	2023
IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)	2023–2026
Conference on Neural Information Processing Systems (NeurIPS)	2023, 2025–2026
International Conference on Learning Representations (ICLR)	2024–2026
International Conference on Machine Learning (ICML)	2024–2026
European Conference on Computer Vision (ECCV)	2024
AAAI Conference on Artificial Intelligence (AAAI)	2025–2026

Teaching Experience

Logic Circuit Design (CSI2111) – Teaching Assistant	Fall 2020, Fall 2021
Multi-core and GPU Programming (CSI4119) – Teaching Assistant	Spring 2021

Skills

Programming	Python, Java, C, C++
Libraries & tools	PyTorch, NumPy, ONNX Runtime, Jupyter Notebook, Matplotlib, pandas
Collaboration	Jira, GitHub workflows
Infrastructure	Docker-based environment setup and deployment
Languages	Korean (native proficiency); English (professional working proficiency)